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In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_squared_error, r2_score
```

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In [2]: data = fetch_california_housing(as_frame=True)
df = pd.concat([data.data, data.target.rename("HousePrice")], axis=1)
df.head()
```

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Out[2]:
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	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	HousePrice
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

```
In [3]: X = df.drop("HousePrice", axis=1)
y = df["HousePrice"]
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In [4]: scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
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In [5]: X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
```

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In [6]: models = {
    "Linear Regression": LinearRegression(),
    "Ridge Regression": Ridge(alpha=1.0),
```

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    "Decision Tree": DecisionTreeRegressor(max_depth=5)
}
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In [7]: from sklearn.metrics import root_mean_squared_error, r2_score
        results = {}

        for name, model in models.items():
            model.fit(X_train, y_train)
            predictions = model.predict(X_test)
            rmse = root_mean_squared_error(y_test, predictions)
            r2 = r2_score(y_test, predictions)
            results[name] = {
                "RMSE": rmse,
                "R2 Score": r2
            }

        results_df = pd.DataFrame(results).T
        results_df
```

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Out[7]:
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	RMSE	R2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

```
In [8]: best_model = LinearRegression() # Example selection
        best_model.fit(X_train, y_train)
        y_pred = best_model.predict(X_test)
        plt.figure(figsize=(6,6))
        plt.scatter(y_test, y_pred, alpha=0.4)
        plt.xlabel("Actual House Prices")
        plt.ylabel("Predicted House Prices")
        plt.title("Actual vs Predicted House Prices")
        plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], color="red")
        plt.show()
```



In []: